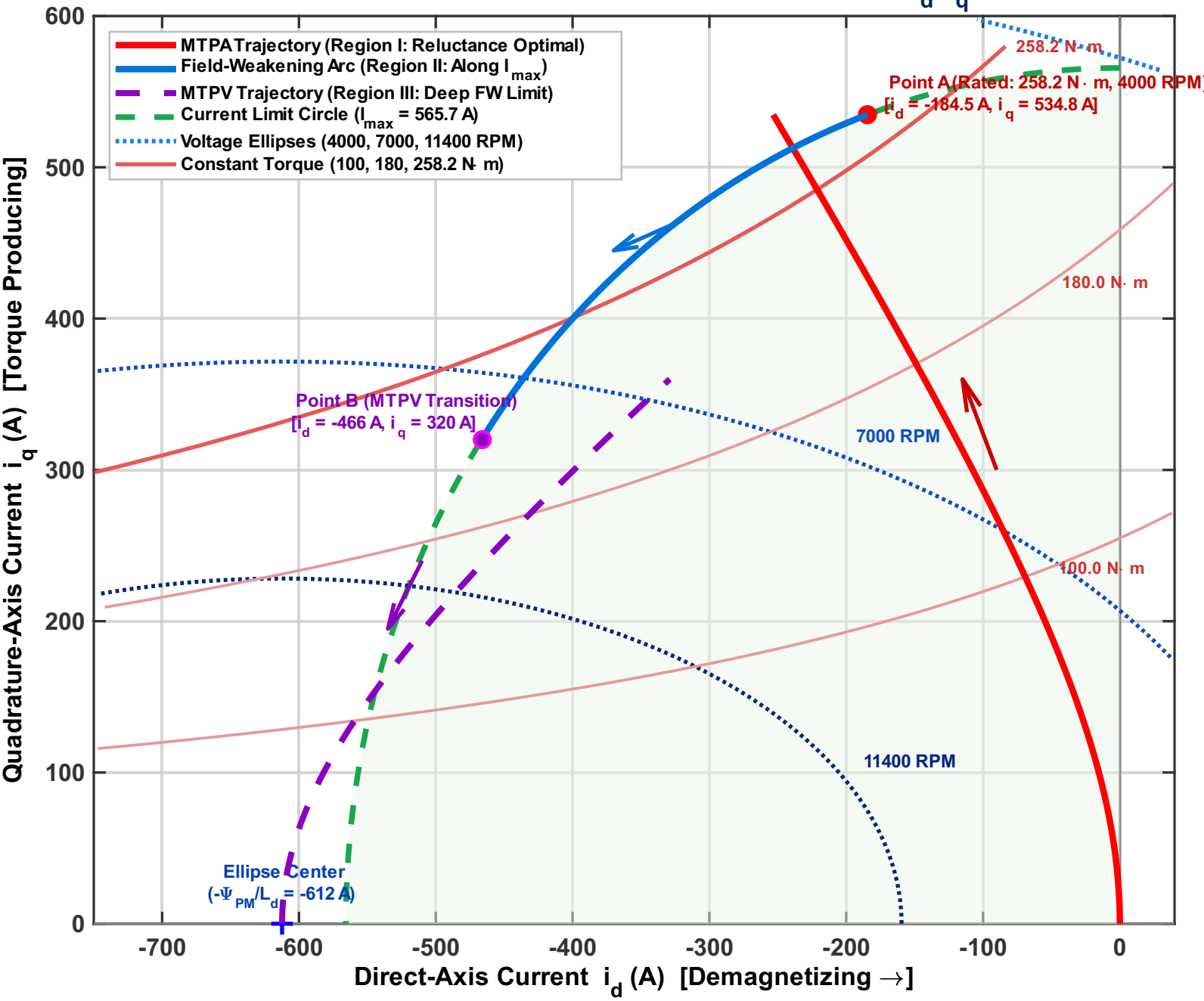


MTPA & MTPV Vector Control Operating Envelope in (i_d , i_q) Plane



OPERATING REGIMES

Region I: MTPA Mode

- Speed: 0 $\rightarrow$ 4000 RPM
- Max torque per ampere
- Reluctance torque boost
- Up to Rated Point A

Region II: Field-Weakening

- Speed: 4000 $\rightarrow$ 8500 RPM
- Along $I_{\max}$ current limit
- Demagnetizing i_d bucks Back-EMF
- Point A $\rightarrow$ Point B

Region III: MTPV Limit

- Speed: 8500 $\rightarrow$ 11400 RPM
- Max torque per volt
- Moves inside circle ($I_s < I_{\max}$)
- High-speed critical limit